

Amar Dhillon

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EDUCATION

University of Maryland, College Park
B.S. in Computer Science, Minor in Statistics; GPA: 3.5

College Park, MD
May 2028

CERTIFICATIONS

Amazon Web Services (AWS) Certified Developer - Associate
Amazon Web Services

Nov 2024 – Nov 2027
Digital Badge

EXPERIENCE

Software Development Engineer Intern
Amazon Web Services

Sep 2026 – Dec 2026
Redmond, WA

- Writing Java policies that set upper/lower bounds on the EC2 capacity Amazon Redshift reserves for patching, backtested to balance availability against cost

Software Engineering Intern
Capital One

Jun 2026 – Aug 2026
Richmond, VA

- Rebuilt the legacy UI of a test-data workflow builder in React, Electron, and TypeScript, adding Jest-tested workflow validation and a React-Flow editor; preferred by **90%** of 30+ surveyed developers
- Built a precomputed tree-style workflow search index, the Claude skill's main search tool, distilling popular workflows from a **270K+ row** PostgreSQL table via Python/SQLAlchemy and Redis within DB permission and rate limits
- Cut workflow generation time **65%** to under 15s with a Claude skill using sub-skill decomposition and Python search/validation tools, guided by CloudWatch + OpenSearch latency logs

Software Engineer (Contract)
Amazon (via UMD App Development Contracting)

Jan 2026 – Present
College Park, MD

- Shipped an Amazon Leo topic-scoring research tool showing public-policy staff which space topics are popular, on Docker + AWS Fargate via GitHub Actions, with AWS CDK (TypeScript) production infrastructure
- Cut end-to-end latency **34%** on a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda, exposed to the app via API Gateway
- Automated production fine-tuning of an AWS Bedrock document-importance model with EventBridge cron triggers and an LLM-as-judge gate that skips redundant retraining; added X-Ray tracing and SNS failure alerts

Machine Learning Research Intern
UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present
College Park, MD

- Developing VLM-based robot grasp selection for PhysTwin, a physics-based digital twin, with a PhD team targeting ICLR
- Engineered a Python pipeline rendering AnyGrasp multi-camera grasp candidates as Open3D overlays for a Qwen3-VL-8B vision-language model, and tuned its prompts to pick grasps that avoid hitting the table
- Reduced research-environment setup time **30%** with uv dependency pinning and Docker, plus psutil time/memory profiling and Pytest suites to locate bottlenecks

Co-Founder, Software Engineer
TerpLabs

Jun 2025 – Present
College Park, MD

- Co-founded and scaled a **110+ member** campus product studio shipping 5 live products, run as agile teams with Jira and GitHub
- Developed Prisma-based SQL querying with Redis query caching for Tortuga, a class scheduler, taking load off compute so it handled **20,000+ visits** in its first 30 days without crashing
- Building a RAG meal-suggestion feature for the official UMD app with GPT-4o mini over ChromaDB + SentenceBERT recipe search; **93%** meal satisfaction among internal testers (in progress)

PROJECTS

IEX Order Book & Matching Engine | *C++17, STL, POSIX, Linux perf*

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine that replayed **50.9M** real IEX messages with no resync, matching 99,122 of 99,123 fills and 100% of top-of-book prices
- Sustained **1.26M** events/sec end-to-end with std::map price levels of FIFO queues and an order-ID hash map for O(1) cancels
- Benchmarked B+ tree vs LSM-tree storage for simulation logs on identical 2M-record workloads; the B+ tree gave **12.8x** read throughput and 22x faster range queries for 27% more disk

TECHNICAL SKILLS

Languages: Python, Java, C++, C, TypeScript/JavaScript, SQL

AWS: Lambda, Kinesis, Fargate/ECS, API Gateway, EventBridge, Bedrock, CDK, CloudWatch, X-Ray, SNS, IAM

Backend & Data: Node.js, React, Electron, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

AI/ML & Tools: PyTorch, Hugging Face, RAG, Open3D, Docker, GitHub Actions, Linux, Pytest, Jest, uv, perf